HOUSEHOLD CARBON FOOTPRINT AND TARGET TRACKER

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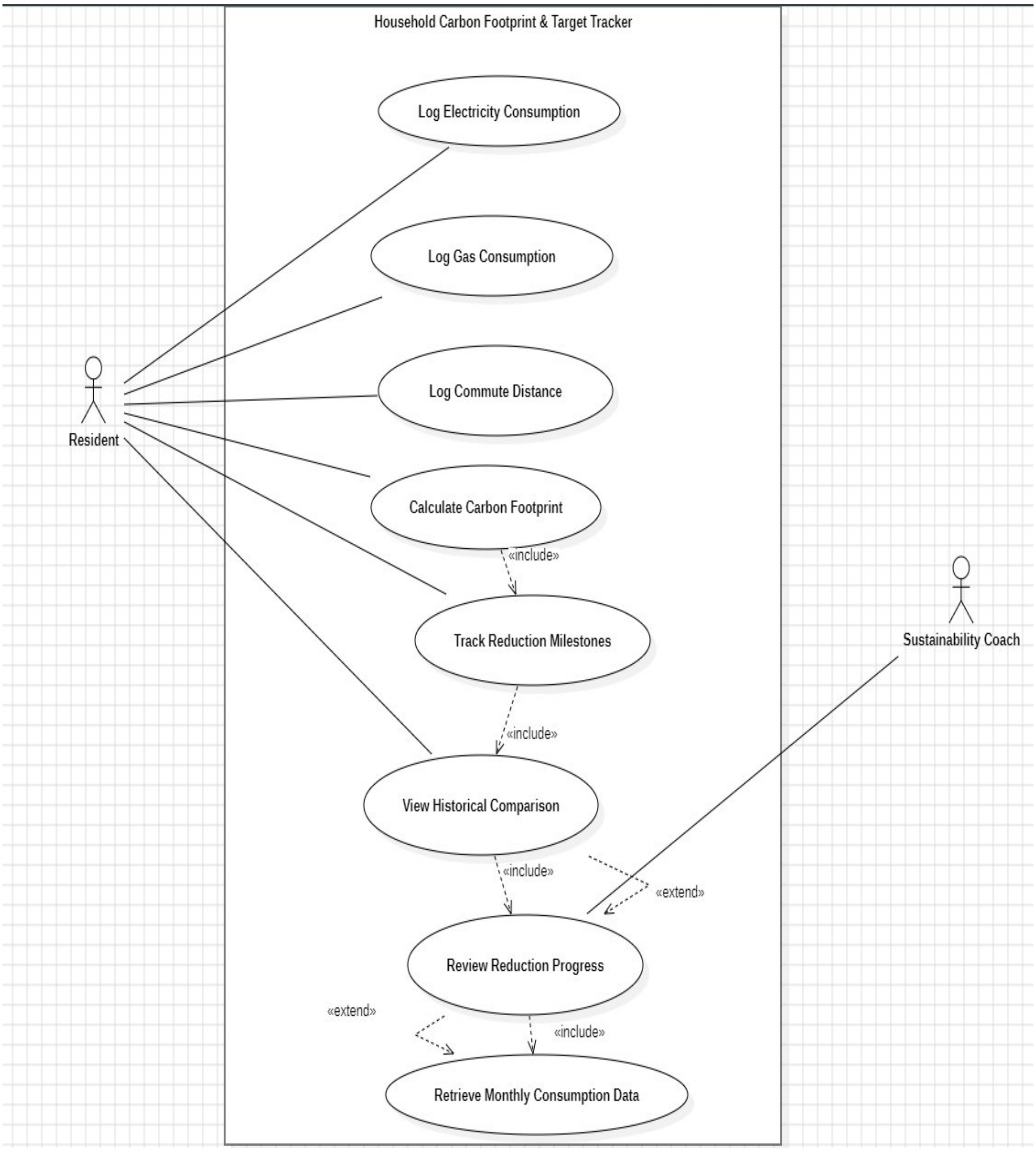
SECTION:H

1. Requirement Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly CO_2e emissions.	High	**Pass:** Calculated emissions match the specified conversion-factor formulas. **Fail:** Zero or negative emission values are generated.	Core carbon-footprint calculation.	Specify the exact conversion factors to be used for electricity and transport.
FR-002	Functional	The system shall allow the Resident to enter monthly household gas consumption.	High	Pass: Valid monthly gas value is stored. Fail: Missing, invalid, or negative value is accepted.	Gas is a household carbon-footprint input.	Specify the unit of gas consumption and acceptable input range.
FR-003	Functional	The system shall store the Resident's monthly electricity, gas, and transport consumption data.	High	Pass: Monthly values can be retrieved. Fail: Submitted data is missing or associated with the wrong month.	Required for historical tracking.	Specify the month format and ensure data is stored under the correct month.
FR-004	Functional	The system shall display the Resident's calculated	High	Pass: Monthly CO_2e value is displayed correctly. Fail:	Allows Resident to understand emissions	Specify the format and

		monthly carbon footprint.		Value is missing or incorrect.		unit in which the monthly CO_2e value is displayed.
FR-0 05	Functional	The system shall track the Resident's progress toward carbon-reduction milestones	High	Pass: Progress is updated after a new result is recorded. Fail: Progress is not updated.	Supports reduction-target tracking.	Define the reduction target or milestone value used to calculate progress.
NFR-001	Nonfunc.	The system shall render the annual carbon reduction milestone chart and interactive historical comparison graphs in under 200 ms.	High	Pass: Chart renders in <200 ms. Fail: ≥200 ms.	Ensures responsive performance.	Specify the test environment and load conditions for measuring the 200 ms response time.
NFR-002	Nonfunc.	The system shall protect household consumption and carbon-footprint data from unauthorized access.	High	Pass: Unauthorized users cannot access another Resident's data. Fail: Unauthorized data is accessible	Protects household data.	Specify the authorization mechanism used to prevent unauthorized access.

USE CASE MODEL



Use-Case Flow Specification

Use Case: Calculate Carbon Footprint

Primary Actor: Resident

Preconditions

1. The Resident has entered the required monthly household consumption data.
2. Monthly electricity (kWh) and transport (km) logs are available for calculation.

Postconditions

1. The system calculates the Resident's total monthly CO_2e emissions.
2. The calculated carbon-footprint result is stored and made available for tracking.

Main Success Scenario

1. Resident selects the **"Calculate Carbon Footprint"** option.
2. System retrieves the Resident's monthly electricity and transport consumption logs.
3. System validates the entered consumption values.
4. System applies the specified conversion-factor formulas to the consumption data.
5. System calculates the total monthly carbon equivalent (CO_2e) emissions.
6. System checks that the calculated emission value is valid.
7. System stores the calculated monthly CO_2e result.
8. System displays the total monthly carbon footprint to the Resident.
9. The carbon-footprint result is made available for reduction-milestone tracking.
10. Use case ends successfully.

Alternate Flow

6a. Invalid Emission Value

- **6a1.** System detects a zero or negative calculated emission value.
- **6a2.** System does not accept the invalid result.
- **6a3.** System informs the Resident that the carbon-footprint calculation could not be completed with the current data.
- **6a4.** Resident is prompted to review and correct the consumption data.
- **6a5.** System resumes the calculation after valid data is provided.